



FIRST SEMESTER 2026 – 2027
Course Handout

Date: 31.07.2026

In addition to Part-I (general handout for all courses in the timetable), this handout provides specific details regarding the course.

Course No.: ME F217
Course Title: Applied Thermodynamics
Instructor-in-charge: Prof. Suvanjan Bhattacharyya
Co-Instructors: Prof. Chennu Ranganayakulu

Scope: The subject matter in this course covers the applications of thermodynamics. This course is designed to acquaint the students with the thermodynamics of power generating and power absorbing machines. The course discusses gas and vapour power cycles, boilers, combined cycle power generation, vapour compression and absorption refrigeration cycles, gas mixtures, psychrometry, and air-conditioning, building cooling/heating load estimation.

Text Books:

1. Çengel Y.A. and Boles M.A., *Thermodynamics*; 8th Ed., 2015; McGraw-Hill Education.

Reference Books:

1. Nag P.K., *Basic and Applied Thermodynamics*, McGraw-Hill Education, 2nd Ed., 2009
2. Nag P.K., *Power Plant Engineering*, McGraw-Hill Education, 4th ed., 2014.
3. Rayner Joel, *Basic Engineering Thermodynamics*, Person Education.
4. Arora RC, *Refrigeration & Air conditioning*, 1st Ed., Prentice Hall India, 2010.
5. Arora CP, *Refrigeration and Air Conditioning*; 3rd Ed. 2009; McGraw-Hill Education.
6. P N Anathanarayanan, *Basic Refrigeration and Air Conditioning*, 4th Ed., 2013, McGraw-Hill Education



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Course Objectives:

- To determine the power output and evaluate the performance of simple thermodynamic cycles for power generation technologies in terms of thermal efficiency.
- To analyze and evaluate the performance of coal based or gas based power generations technologies.
- To analyze and evaluate the performance of vapor compression and vapor absorption refrigeration cycle.
- To analyze and evaluate gas turbine cycle with multistage compression with intercooling, multistage expansion with reheat and regeneration terms of its efficiency.
- To estimate air conditioning requirements of a space using various air-conditioning processes.

Course Plan

Module 1: Vapor Power Cycles

Learning Objectives

- To analyze ideal Rankine cycle for a vapor power plant using T-s plot and evaluate thermal efficiency by applying first law and second law of thermodynamics.
- To evaluate performance of Rankine cycle considering isentropic efficiencies of turbine and pump.
- To represent the Rankine cycle processes on the Mollier diagram (enthalpy-entropy chart).
- To analyze reheat Rankine cycle for a vapor power plant using T-s plot and evaluate its performance by applying first law of thermodynamics.
- To evaluate performance of regenerative Rankine cycle using feed water heaters.
- To study boilers their mountings and accessories.

Learning Strategy:

Topics	Text book Chap/Sec
Carnot cycle deficiencies, Simple vapor power cycle, Rankine Cycle, Actual vapor power cycle, Mean temperature of heat addition. Techniques for efficiency improvement, Reheat and Regenerative cycles with open & closed feed water heaters. Combined cycle power plant and its thermodynamics. Combined Heat and Power.	10, TB1





Module 2: Refrigeration Cycles

Learning Objectives

- To study simple refrigerators and heat pumps systems and the evaluate of their performance.
- Analyze the ideal and actual vapor compression refrigeration cycle.
- Evaluate the second law efficiency of vapor compression refrigeration cycle.
- Introduce the concepts of vapor absorption-refrigeration systems.
- To determine maximum COP of an ideal absorption refrigeration system.

Learning Strategy:

Topics	Text book Chap/Sec
Vapor Compression Refrigeration Cycle, Actual Vapor Compression Cycle, HeatPump, Second Law Efficiency of Vapor Compression Cycle.	11, TB1
Basic principle of a Simple Vapour–Absorption System, Comparison of vapour compression refrigeration systems with continuous vapour absorption refrigeration systems, maximum COP of an ideal absorption refrigeration system, properties of ideal and real refrigerant-absorbent mixtures, single stage vapour absorption refrigeration system with solution heat exchanger. Introduction to compressors.	11; RB3 & RB4

Module 3: Psychrometry

Learning Objectives

- Differentiate between dry air and atmospheric air.
- To define and calculate the specific and relative humidity of atmospheric air.
- To calculate the dew-point temperature of atmospheric air and to relate the adiabatic saturation temperature and wet-bulb temperatures of atmospheric air.
- To determine the properties of atmospheric air using psychrometric chart.
- To apply the principles of the conservation of mass and energy to various air-conditioning processes.
- To estimate heating and cooling loads of a space to be air conditioned.

Learning Strategy:

Topics	Text book Chap/Sec
Psychrometric Properties, Psychrometric Chart, Mixing of moist air, Psychrometry of air conditioning processes.	14, TB1; RB3 & RB4
Inside and Outside Design Conditions, Simple Summer Air Conditioning System, Supply Air condition and flow rate, Apparatus Dew Point, Winter Air Conditioning. Building Cooling Load and Heating Load Estimate.	13 & 14, TB1; 30, RB3 & RB4



Module 4: Gas Power Cycles

Learning Objectives

- To analyze Gas power cycles for a perfect gas using $P-v$ and $T-s$ plot and evaluate its performance using first law of thermodynamics.
- To evaluate and compare the thermal efficiency of regenerative Stirling/Ericsson cycle with that of Carnot cycle.
- To evaluate effects of multistage compression with intercooling, multistage expansion with reheat and regeneration on performance on a simple gas turbine cycle in terms of its efficiency.
- To analyze combined cycle and binary cycle power plants using first and second law of thermodynamics and evaluate their thermal efficiency.

Learning Strategy:

Topics	Text book Chap/Sec
General introduction to Subject, Simple steam power plant, Internal Combustion Engines, Gas turbine plants, Refrigeration cycle and COP. Carnot Cycle, Air Standard Cycles, Otto Cycle, Diesel Cycle, Dual Cycle. Stirling cycle, Ericsson Cycle, Brayton cycle: intercooling, reheating and regeneration. Jet propulsion.	Class notes 9, TB1

Module 5: Thermodynamic Relations

Learning Objectives

- To understand the Maxwell relations
- To discuss the Clapeyron equation and understand its utility
- Understanding the general thermodynamic relations and Joule Thomson effect.

Learning Strategy:

Topics	Text book Chap/Sec
Maxwell relations, Clapeyron Equation, Joule Thomson Coefficient and General Thermodynamic Relations.	12, TB1



Evaluation Scheme:

Components	Duration	Weightage (%)	Maximum Marks [200]	Date & Time	Remarks
Quizzes (Best 3 out of 4)	45 min.	15	30	To be announced in class	Closed book
Mid-semester test	90 min.	25	50	07/10/2026	Closed book
Lab Report, Quiz and Lab Examination	-----	25	50	To be announced in class	-
Comprehensive test	120 min.	35	70	08/12/2026	Open book

Mid-semester grading: It will be announced normally in the month of November.

The best 3 out of 4 quizzes will be considered for grading.

The lab will cover experiments related to Applied thermodynamics. There will be separate marks awarded to lab reports for each experiment, comprehensive viva and quizzes will be conducted for the lab along with the final lab examination.

Chamber Consultation Hours:

Tuesday, 6-7 pm.

Notices: All notices related to this course will be put on the Nalanda.

Make-up Policy: It will be as per the institute rules. Only genuine cases will be considered. No makeup will be provided for the tutorial and lab quizzes.

Instructor-in-charge

Prof. Suvanjan Bhattacharyya
ME F217